Roll No. | | | | | | | | | | |

Total No. of Pages : 02

Total No. of Questions : 13

B.Pharma (Sem.-4)

PHARMACEUTICAL ORGANIC CHEMISTRY-III

Subject Code : BP-401 T

M Code : 75843

Date of Examination: 11-12-2024

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.**

SECTION-A

1. **Write short notes on:**

   a) Describe the conditions that lead to a molecule being chiral or achiral.

   b) Can you provide an example of a molecule with both E and Z stereoisomers?

   c) What is a meso compound, and how does it differ from other chiral molecules?

   d) Identify the key factors that lead to the formation of geometrical isomers in organic compounds.

   e) What is meant by the term "racemic modification" in stereochemistry?

   f) Which compound among pyrrole, furan, and thiophene exhibits the highest aromaticity?

   g) Compare the structural differences between pyridine, quinoline and isoquinoline.

   h) Write structures and specific medicinal uses of imidazole.

   i) What is the primary function of Wolff-Kishner reduction in organic synthesis?

   j) What is the purpose of the Oppenauer oxidation in synthetic chemistry?

SECTION-B

2. Discuss the RS system of nomenclature for optical isomers. How does it relate to absolute configuration?

3. Provide a comprehensive overview of the Claisen-Schmidt condensation, explaining the reaction mechanism.

4. Compare and contrast the conformational isomerism in ethane, n-butane and cyclohexane.

SECTION-C

5. What is racemic modification, and how does it relate to the resolution of racemic mixtures?

6. Explain the principles of asymmetric synthesis, including partial and absolute asymmetric synthesis.

7. Define atropisomerism and describe the stereochemistry of biphenyl compounds.

8. Define stereospecific and stereoselective reactions in organic chemistry.

9. Discuss the synthesis and medicinal uses of pyrrole and thiophene derivatives.

10. Explain the synthesis of azepines and discuss their potential therapeutic applications.

11. Explain the concept of basicity in pyridine and compare it to other heterocyclic compounds.

12. Write the synthesis and medicinal uses of indole and its derivatives.

13. Discuss Beckmann's rearrangement and its role in converting oximes into amides.

NOTE : Disclosure of Identity by writing Mobile No. or Marking of passing request on any paper of Answer Sheet will lead to UMC against the Student.

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